Project Proposal

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```
library(tidyverse)
library(tidymodels)
library(readr)
library(vroom)
library(patchwork)
library(knitr)
```

```
# load dataset

# vroom function from:
# https://cran.r-project.org/web/packages/vroom/vroom.pdf#:~:text%20#%20
# =include_na%20If%20TRUE%20and%20the%20data%20contains
# ,A%20format%20%20#%20specification%20If%20set%20to%20%22%22:

food_access <- vroom("data/Food Access Research Atlas.csv",
                     na = c("", "NA", "NULL"),
                     show_col_types = FALSE)

# load data dictionary to paste into data README.md
data_dict <- read_csv("data/VariableLookup.csv", show_col_types = FALSE)
```

Introduction

There are 18.8 million Americans living in food deserts today.) — areas marked by limited access to affordable and nutritious food. The USDA has defined food deserts to be census tracts that have a poverty rate of 20% or greater (or median family income at the most 80% of statewide family income) and where at least 33% of the population (or at least 500 people) live more than 1 mile from the supermarket in an urban area/ 10 miles in a rural area. Food deserts are created not only due to distance from healthy food sources but also due to structural decisions that have historically limited healthy food access for minority communities.

Understanding this issue requires examining how redlining policies institutionalized neighborhood disinvestment, disproportionately exposing low-income and minority communities to less healthy food environments. These disparities carry serious consequences: food deserts are linked to worse health outcomes, including shorter life expectancy and a higher risk for adverse cardiovascular events.). Given these health risks and the disproportionate racial impact of food deserts, it is critical to evaluate what factors contribute to their persistence and how current policy can work toward reversing years of racial residential segregation.

This issue is particularly relevant to us as Duke students, as 9 of the top 10 states with the greatest share of residents living in food deserts are in the South. This proximity makes understanding the structural drivers of food deserts a matter of community concern. Thus, I seek to answer the following question: in what ways do the poverty rate of a census tract, the number of households receiving SNAP benefits, the number of households without a vehicle, and urban/rural classification influence how many low-income people in a census tract live in a low food access area?

I hypothesize that poverty rate and receiving SNAP benefits will be the strongest, positive predictors of food desert prevalence, as economic disadvantage is the most consistent driver of limited food access in the literature. Furthermore, SNAP receipt and food desert classification are likely spatially linked. A number of SNAP recipients live in areas that were historically redlined, and since redlining similarly concentrated poverty and limited neighborhood investment, I hypothesize that these two conditions tend to occur in the same communities. Since my response variable measures the population count of those living in a food desert by census tract, I expect population size to naturally influence this measurement. For this reason, I predict that urban/rural classification will play a meaningful role in mirroring that relationship. Finally, I am interested in exploring the relationship between vehicle access and food desert prevalence. Since living in a food desert is defined in part by distance from a supermarket, lacking a vehicle would make it harder to bridge that gap. However, I do not expect this to be a strong predictor, as the absence of a vehicle does not necessarily mean the absence of transportation access (public transport could be an alternative in some tracts). Since public transportation is not captured in this data set, I will acknowledge it as a limitation and interpret the vehicle access coefficient with that caveat in mind.

Data description

The 2019 Food Access Research Atlas that I will be analyzing is from the USDA Economic Research Service. It provides an overview of food access indicators in the US by measuring variables like supermarket accessibility, race, poverty rate, and median income according to census tracts. This data set is a combination of data collected by the 2010 Decennial Census, the 2014-18 American Community Survey (ACS), and two 2019 lists of supermarkets, one being from Trade Dimensions TDLinx. The 2010 Census provided population data relating to age, race, income, poverty rate, and residence in group quarters. The 2014-18 ACS contributed information about income, vehicle availability, and SNAP participation. Finally, two lists of

grocery stores were combined to produce a list of stores that have nutritious and affordable food. This data was then used to compute the variables that communicate the distance that one lives from a grocery store.

Each observation in the data represents an individual census tract measured by the 2010 US Census and uniquely identified by the `CensusTract` variable. Generally, this data set measures food store access for individuals and for neighborhoods, using different variables to measure low-income and low-access communities that lack access to healthy food sources. Most of the measures include measures of accessibility to sources of healthy food (by distance to a store), individual resources that impact accessibility (vehicle availability or income), and neighborhood-level indicators of resources (eg, median income).

Data processing

This data set currently has 72,531 observations representing the number of Census tracts in the 2010 U.S. Census. Training a regression model on too many observations, however, poses a risk of overfitting. The model has the potential to learn random noise and thereby ignore the true pattern underlying the data. It can also make statistical significance tests too sensitive, resulting in "statistically significant" differences when there were likely not any. To address this issue, I will take a random sample of 500 observations from the data set. This smaller sample via randomization also seeks to avoid the issue of spatial dependence, since census tracts near each other are likely to have similar data.

In addition to reducing the size of my data set, I also plan to convert the indicator predictor variables of interest to categorical variables. Currently, the data has marked each variable except `State` and `County` as numeric. However, variables like `Urban`, which simply display if a tract is urban or not, should be represented as a factor with two levels. Once I convert these variables into categorical predictors, my regression model and data visualizations will be interpretable.

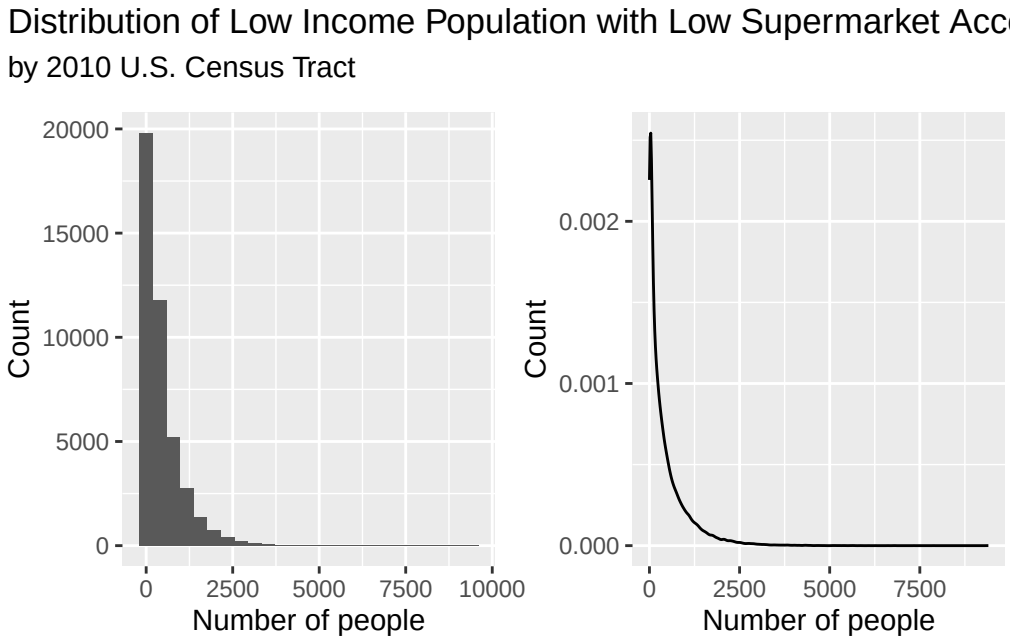
Finally, I will address the issue of missing data. There are 29,957 missing observations in the `LALOWI1_10` response variable. To address this, I will remove all rows containing missing values from the dataset. I chose this approach over replacing missing values with 0 because, in the context of this variable, a value of 0 would imply that no low-income residents in a given census tract live far from a supermarket. This scenario is unlikely and would misrepresent the true data. It is more plausible that these tracts were not sampled or did not meet the criteria for measurement, rather than having zero qualifying residents. Additionally, imputing 0 for nearly 30,000 observations would artificially concentrate the distribution at zero and distort the model's estimates.

Analysis approach

```
# summary statistics
summary <- food_access |>
  summarise(
    min = min(LALOWI1_10, na.rm = TRUE),
    Q1 = quantile(LALOWI1_10, 0.25, na.rm=TRUE),
    median = median(LALOWI1_10, na.rm = TRUE),
    mean = mean(LALOWI1_10, na.rm = TRUE),
    Q3 = quantile(LALOWI1_10, 0.75, na.rm=TRUE),
    sd = sd(LALOWI1_10, na.rm = TRUE),
    IQR = Q3 - Q1,
    max = max(LALOWI1_10, na.rm = TRUE),
    # following line from https://stackoverflow.com/questions/29076536/
    # how-can-i-count-number-of-na-values-in-dataset
    n_missing = sum(is.na(LALOWI1_10))
  )

# visualize distribution of response var LALOWI1_10 - ind food access
p1 <- food_access |>
  ggplot(aes(x=LALOWI1_10)) +
  geom_histogram(bins=25) +
  labs(x = "Number of people", y = "Count")

p2 <- food_access |>
  ggplot(aes(x=LALOWI1_10)) +
  geom_density() +
  labs(x = "Number of people", y = "Count")

summary
```

```
# A tibble: 1 x 9
    min    Q1 median  mean    Q3    sd   IQR   max n_missing
  <dbl> <dbl>  <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>     <int>
1     0    49    228  442.   605  589.   556  9402     29957
```

```
plots <- p1 + p2
plots + plot_annotation(
  title = "Distribution of Low Income Population with Low Supermarket Access",
  subtitle = "by 2010 U.S. Census Tract"
)
```



The response variable `LALOWI1_10` represents the "low-income population count beyond 1 mile for urban areas or 10 miles for rural areas from [the] supermarket" (data dictionary). Since the USDA definition for food deserts meets this criterion, I plan on using `LALOWI1_10` to represent the number of people in a census tract living in a food desert.

The histogram displays a unimodal, right-skewed distribution, indicating that most census tracts have relatively small low-income populations living far from a supermarket. The long right tail suggests that while a small number of tracts have very high counts, the majority are concentrated at lower values. Due to the skew, the mean is not an accurate indicator of the center of the distribution. Thus, the median of 228 suggests that in a typical census tract, about 228 low-income residents live beyond 1 mile (urban) or 10 miles (rural) from a supermarket. The spread of the data is about 556 and most of the data (middle 50%) is between 49 people and 605 people living in a food desert in a census tract. There is a minimum of 0 people and a maximum of 9402 people in a census tract living in a food desert. There are a couple of outliers in the data at census tracts with more than 4000 low-income people living with little access to healthy food. Though the histogram doesn't show these outliers, we can look to the density plot and the summary statistics to understand that these outliers exist outside the bins on the histogram plot.

To analyze the factors that influence the size of the low-income population living far from healthy food (`LALOWI1_10`), I plan to examine `PovertyRate` (poverty rate in a census tract), `Urban` (if the tract is in an urban or rural area), `TractSNAP` (number of housing units receiving SNAP benefits in a tract), and`TractHUNV` (total number of housing units without a vehicle). Since my response variable is quantitative, I will fit a multiple linear regression model to understand

how the predictors are associated with a difference in the number of low-income people living far from a supermarket in a census tract. These predictors were selected because literature shows that food deserts are characterized by factors like higher poverty, small populations, lower educational attainment, and less transportation access, to name a few. Informed by these findings, I have selected predictors like `PovertyRate` and `TractHUNV` . I chose to add `TractSNAP` because I am interested in seeing if receiving SNAP benefits is a good proxy for food insecurity. Additionally, many studies analyzed how the minority population influenced the population in a food desert. I chose `Urban` to see if the population density played a role instead. While the data contains many variables that mean the same thing, just measured the same way – for instance, a population amount or proportion – I chose all of the quantitative predictor variables to represent a count of individuals instead of a proportion to be consistent with the way the response variable is measured.

Data dictionary

The data dictionary can be found here.

AI Disclosure

- I asked AI about an error or why code would do something I didn't want.